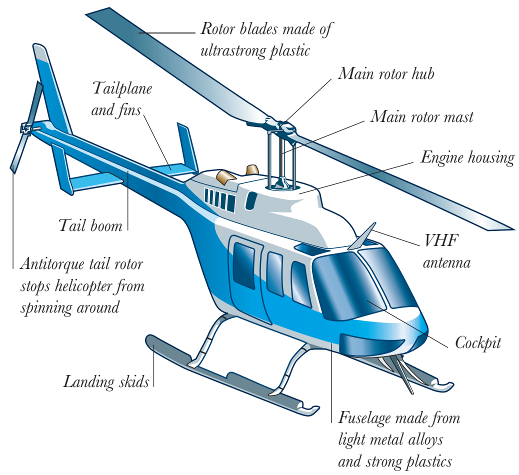
HELICOPTER



FLYING A HELICOPTER

The helicopter is the most versatile aircraft, able to fly forward, backward, or sideways. The helicopter pilot maneuvers by changing the blades' pitch (angle), while the tail rotor stops it from spinning round.

Stealth technology
A combination of geometric shaping and material technologies aimed at producing an aircraft that has low-observability, primarily in relation to enemy radar detection.

STOL
STOL (short takeoff and landing) refers to a system by which aircraft take off and land over a short distance.

Streamline
To shape an object, such as an airfoil or fuselage, so that it creates less drag and moves smoothly through the air.

Subsonic
Slower than the speed of sound.

Supercharger
A type of compressor in a piston engine used to boost power by increasing the amount of air fed to the cylinders.

Supersonic
Faster than the speed of sound.

Swept wing
A wing that it is angled toward the rear of the aircraft, in order to reduce drag.

Swing-wing
Also known as Variable-sweep. *See* WING SHAPES *panel, below.*

Tachometer
The gauge that displays an aircraft's engine speed.

Tailplane
A fixed, horizontal structure attached to the tail or the rear part of the fuselage. Also know as a horizontal stabilizer.

Throttle
The lever that regulates the amount of fuel reaching an engine, which in turn affects the amount of thrust generated.

Thrust
The force, generated by an engine, that pushes an aircraft through the air.

Thrust vectoring
Controlling a jet aircraft's movement by altering the angle of propulsive thrust, via a jet engine's rotating nozzles.

Torque
The twisting force created by a turning component, (such as a propeller) powered by an engine.

Tractor
A configuration in which the propeller is situated in front of the engine or fuselage and pulls the aircraft through the air.

Trailing edge
The rear edge of an airfoil, usually thin and sharp.

Triplane
An aircraft with three sets of wings, arranged one on top of the other.

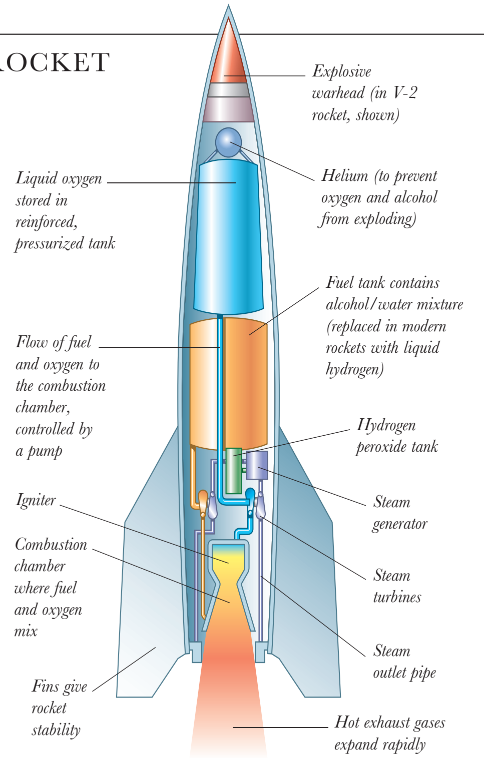
Turbofan
See JET ENGINES *panel, left.*

Turboprop
See JET ENGINES *panel, left.*

Turbulence
Irregular air flow, which can cause an aircraft to fly unpredictably.

Ultralight
A small, powered aircraft, often in the form of a hang-glider.

ROCKET



ROCKET ENGINE

The rocket is the simplest and most powerful kind of engine – a reaction engine. Fuel (solid or liquid) is burned in an open-ended combustion chamber and the resulting heated, high-pressure gases escape at high speed out of the chamber, providing thrust.

Undercarriage
The unit beneath an aircraft – consisting of wheels (or skis or floats), shock absorbers, and support struts – that supports the aircraft on the ground.

Variable-geometry
Also known as Variable-sweep. *See* WING SHAPES *panel, below.*

Variable-sweep
See WING SHAPES *panel, below.*

VTOL
VTOL (vertical takeoff and landing) is a system that allows aircraft to take off and land vertically, eliminating the need for a runway altogether.

Widebody
A type of airliner with a passenger cabin sufficiently wide for it to be divided into three sets of multiple-seat columns.

Wing flap
The movable part of the wing's near trailing edge, which is used to increase lift at slower air speeds and slow the aircraft on landing.

Winglet
A small, upright structure on the wingtip, used to lessen tip vortices and therefore reduce drag.

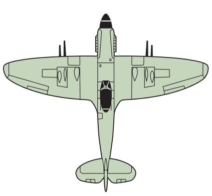
Wing warping
A system invented by the Wright brothers to provide lateral control to their biplane aircraft, by twisting its wingtips (via wires in the wings). It was later replaced by movable, hinged ailerons.

Wire-braced
The term given to any part of an aircraft, such as a wing or fuselage, that uses tensioned wire bracing to maintain rigidity.

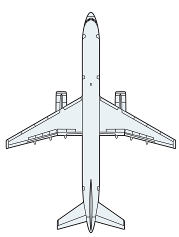
Yaw
See BASIC CONTROLS *panel, p.429.*

Zeppelin
Any of the airships built by the Zeppelin company, founded by Count von Zeppelin (1838–1917).

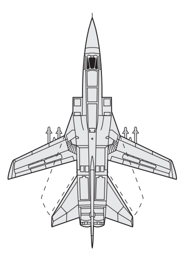
WING SHAPES



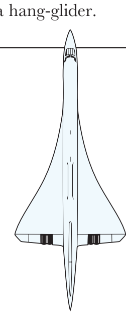
STRAIGHT
Short, straight wings produce good lift and low drag at medium speed. Propellers or jet engines provide the power that produces the lift.



SWEPT
Swept-back wings minimize drag at high speed. However, lift is also reduced, requiring high takeoff and landing speeds.



VARIABLE-SWEEP
Hinged wings can be slanted backward during flight to reduce drag. The wings are at right angles for takeoff and landing to increase lift.



DELTA
Supersonic aircraft often have delta wings to help retain control of the aircraft within the shock wave that forms around it at such high speeds.